Communicating Quantitative Information: Graphs

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In 1978 I was contacted by a small company called Consulting Statisticians, Inc. They had received a request for proposals issued by the National Institute of Education. Someone in the government had apparently noticed that many people could not read charts and graphs very well and wanted to spark more research on the topic. I was intrigued. Then-graduate student Steven Pinker and I wrote a proposal, and it was funded. So began my work on graphs. That work ultimately led to the eight principles that guide the recommendations in this book, and graphs serve as a prototype for how to apply those principles.

Graphs are pictures intended to convey information about numbers and relationships among numbers. Graphs can convey the information in a large set of numbers more effectively than just listing the numbers—or can fail miserably. In this chapter I help you avoid making graphs that confuse rather than illuminate, and instead will guide you to make graphs that could be worth many numbers—becoming pictures that are truly worth 1,000 words.

To Graph or Not to Graph? The 3 Big Questions

The following considerations will help you to decide whether you should use a graph or would be better off with a table or text.

1. Do you want to illustrate relative amounts?

If so, then a graph is appropriate. Graphs can also specify specific amounts (in labels at specific points), but what they do best is to exploit the "more-is-more" rule: We automatically see larger wedges, longer bars, or higher points along a line as signifying greater quantities, and it's easy for us to see relative differences along these visual dimensions.

2. Do you have data needed for a specific purpose?

If your data address a specific issue, then a graph can be illuminating. Decide what questions audience members should be able to answer from the display, and organize the data accordingly. As usual, don't include any more or less information than is needed to answer the relevant questions.

Example: If you want the audience members to know whether sales are increasing at the same rates in different parts of the country, plotting the data separately for each of your product lines would only be an obstacle; audience members would mentally have to average over lots of lines or bars to obtain the information they needed. On the other hand, if you want the audience to know about differences in sales of commercial and personal equipment, it would be an error not to include separate bars or lines for each type. There is no way of mentally breaking down a single point or bar into its constituent parts.

3. Can you use concepts and display formats that are familiar to the audience?

If not, then a graph will not serve you well. I was shocked when I discovered that a substantial number of undergraduates at Harvard University don't really know how to read line graphs. The "more-is-more" rule has been a revelation for some of these students. Don't assume that even an intelligent and presumably well-educated audience will necessarily understand information graphics. If you decide to use a graph:

- Pick a display format that's familiar to a particular audience. When in doubt, explain how the graphic works.
- Use concepts familiar to that audience.
- In general, stick to the tried-and-true, "traditional" graphs that I discuss in this chapter.

In the remaining sections of this chapter, we will consider pie graphs, visual tables, line graphs, bar graphs, step graphs, scatterplots, and multi-panel displays.

Pie Graphs: 6 Tips

The pie graph is the most common format that shows how a whole is divided into parts. The relative area of each component, the wedge, represents the proportion of the whole taken by the component.

1. Use a pie to convey approximate relative amounts.

Yogi Berra once commented that "Baseball is 90 percent mental. The other half is physical." He would never have made this blooper if he had tried to use a pie graph to express this idea. And you'll never make similar gaffes if you display such data in a pie graph, which illustrates proportions of a whole. The main drawback to pie graphs is that it's often difficult to obtain relatively precise amounts from them, such as the percentage of one part. However, this is not a problem if the amount is 25 percent, 50 percent or 75 percent, which are easy to discern at a glance. If other amounts are illustrated, the precise amounts will not be easy to obtain because it is difficult visually for us to measure the angle, chord (i.e., straight line that connects the two sides of the wedge at the point where those lines

meet the surrounding circle—making the wedge into a triangle), or area of each wedge precisely.

Example: It is easy to use the "Do" graph in Figure 5.1 to get a general sense of the relative proportions, but difficult to use the "Don't" graph to determine that there were 13 percent more car and mobile home loans than bank and finance loans.

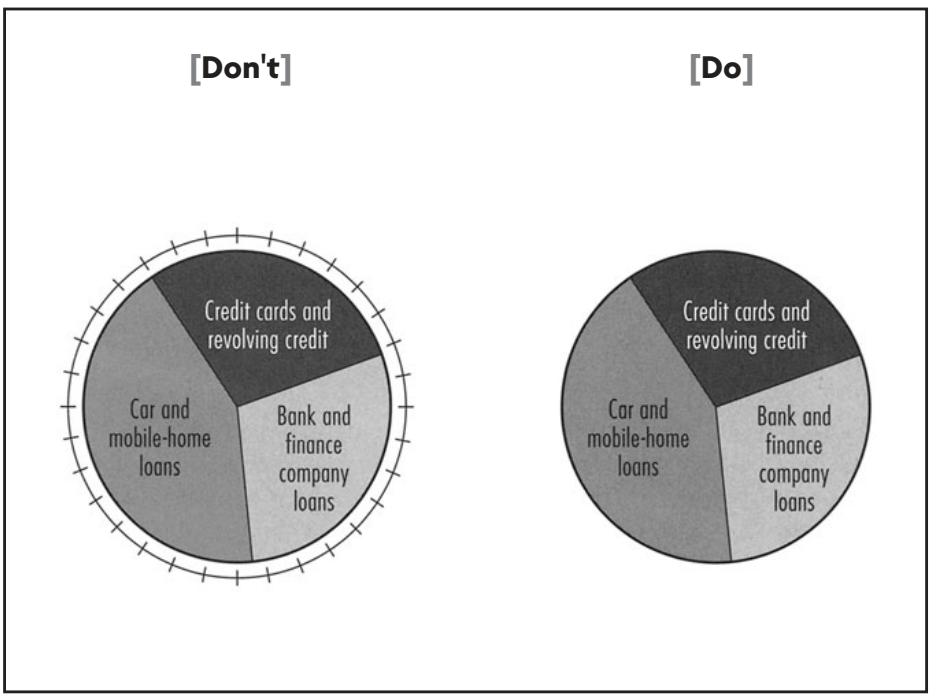


Figure 5.1. Do not use a scale with pie graphs; the viewer will have to struggle to count the number of ticks. In this graph, credit cards and revolving credit are the topic of interest, and hence this information is most salient.

2. Label wedges if precise values are important.

A good way to convey precise amounts while still depicting relative proportions with pie graphs is simply to label the wedges, either by putting the numbers in them (if space permits) or next to them. This converts the pie into a hybrid display, part pie and part table.

3. Use an exploded pie to emphasize a small proportion of parts.

Construct an exploded pie display by displacing the important slice or slices, as if a wedge of pizza had been pulled out from the pie. If the audience is supposed to make only approximate visual comparisons,

the pie format provides a particularly good way to draw attention to a small percentage of the total number of components.

Example: Look at the "Do" graph in Figure 5.2; it is immediately obvious that credit cards and revolving credit are the most important aspects of the display. In contrast, in the "Don't" version it is not clear what the audience should focus on.

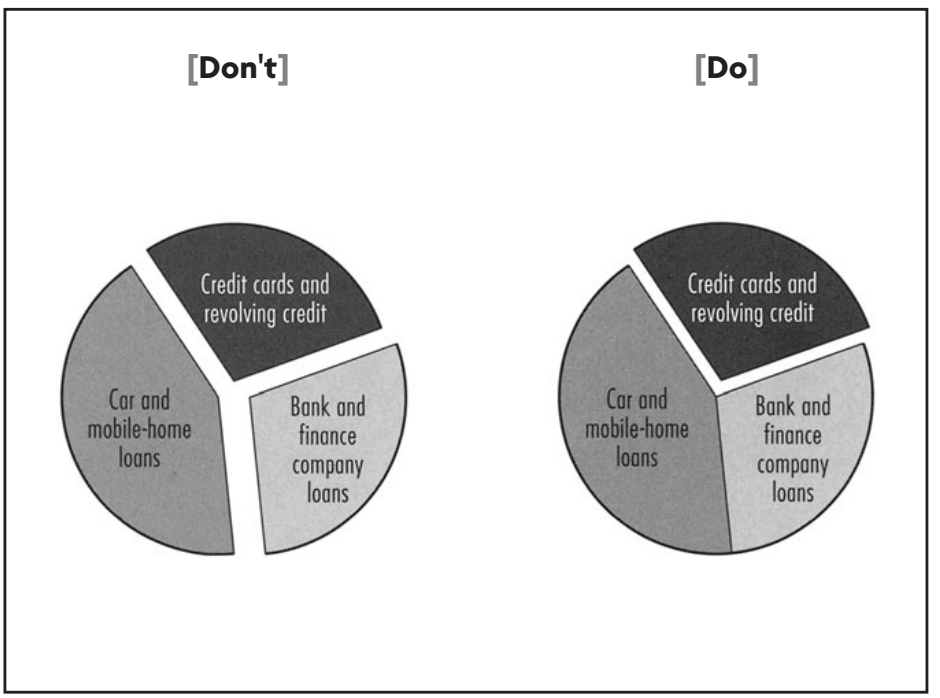


Figure 5.2. Use an exploded pie only when a distinct contour can be disrupted by exploding the wedges; the two intact pieces of the pie on the right define a contour, which makes the exploded wedge stand out.

4. Explode a maximum of 25 percent of the wedges.

If you decide to use an exploded pie, you must decide which part or parts to emphasize. If too many wedges are exploded, as they are in "Don't," of Figure 5.3, the viewers won't know where to look. I once saw a cat freeze in confusion when it knocked over a trash can and some half-dozen mice scattered in different directions; it just didn't seem to know which one to pounce on. Don't make your viewers like that cat. I offer 25 percent as a rough guideline; there is no hard-and-fast percentage. The critical consideration is that enough wedges remain in the pie to make the exploded wedges appear to disrupt the contour of the whole.

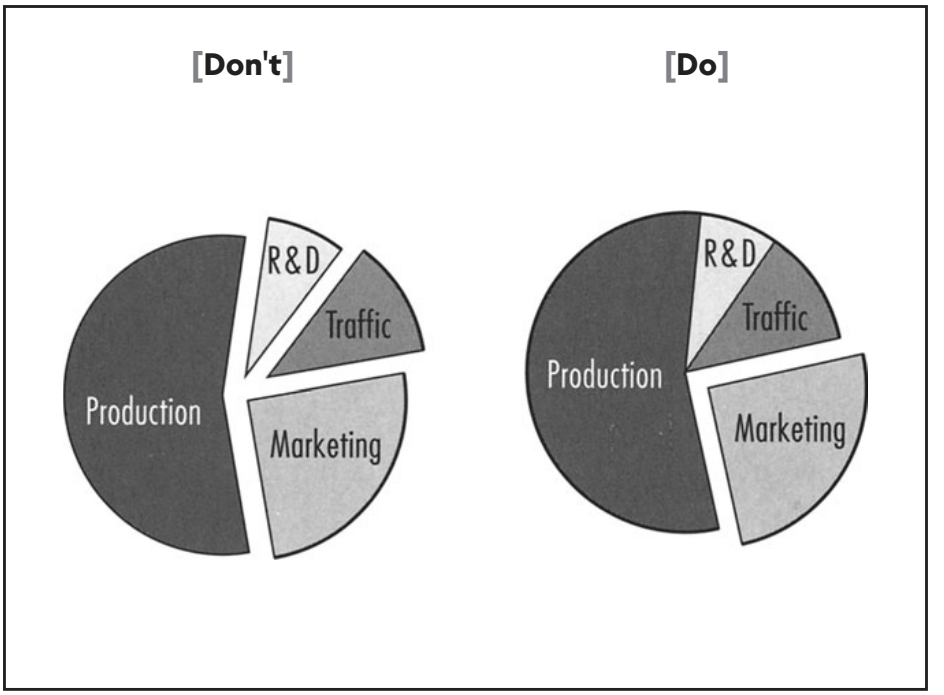


Figure 5.3. If too many wedges are exploded, none stands out.

5. Arrange wedges in a simple progression.

Unless there are reasons to order the wedges in a specific way, it will be easiest to compare the wedges if they are arranged in order of size (see Figure 5.4).

- In general, order from smaller to larger, with size increasing clockwise. Because values on a clock face increase in a clockwise direction, we expect greater quantities to be indicated by greater arcs in a clockwise direction.
- However, if you wish to emphasize the larger components, put the largest at the top or in the "1:00" position (the first that the viewer will focus upon when scanning in a clockwise order), and arrange the other wedges in decreasing size from this anchor point.

6. If proportions vary greatly, don't use multiple pies to compare corresponding parts.

Compare the two pies in the "Don't" version of Figure 5.5. How did Peugeot and Renault do in the European Community (EC) in general

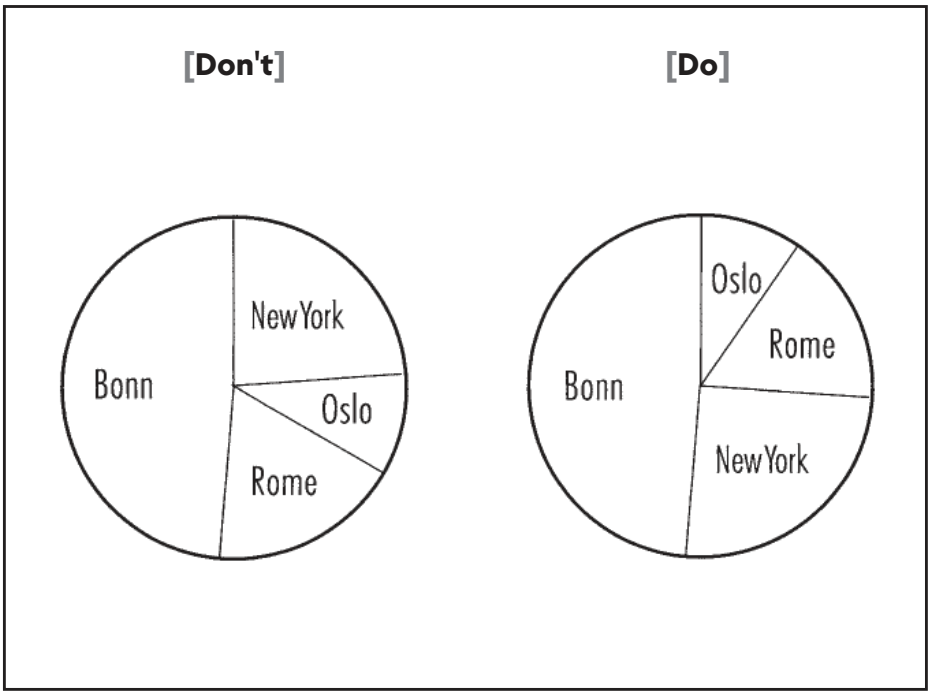


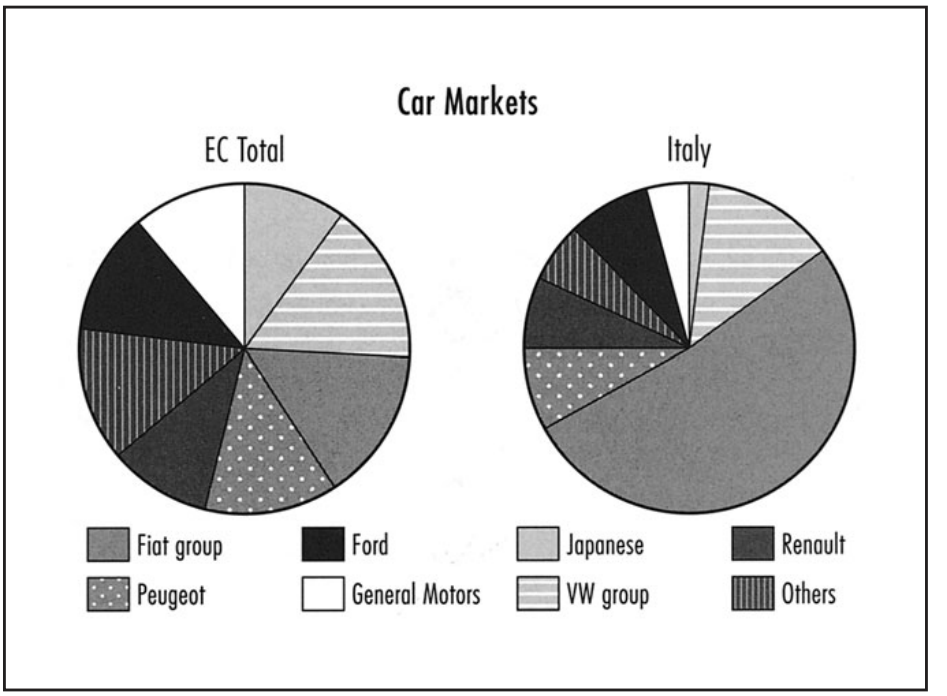
Figure 5.4. In Western culture, readers expect quantities to increase clockwise around a circle.

(left) compared with Italy in particular (right)? (These data were cited in 1991, prior to the formation of the European Union.) This is difficult to fathom, in part because the relevant wedges are in different locations in the two pies—a shift that cannot be avoided when the proportions vary widely.

Now compare the two "Do" pie graphs; it is clear that there is a rough correspondence between the revenues per region (left) and employees per region (right). It is easy to compare multiple pies when the wedges are in roughly the same positions in each. But, as illustrated in "Don't," when the corresponding parts are in different locations, the audience is forced to search for them one at a time—which requires effort. Our brains have limited processing capacity, and if forced to strain, many audience members may simply give up. (Also note that because Western Europe was of greatest interest to the designers, the largest wedge is put at the beginning of the clockwise progression.)

- If the proportions are very different across levels, and the audience is supposed to compare specific components, a bar graph is the preferable format.

[Don't]



[Do]

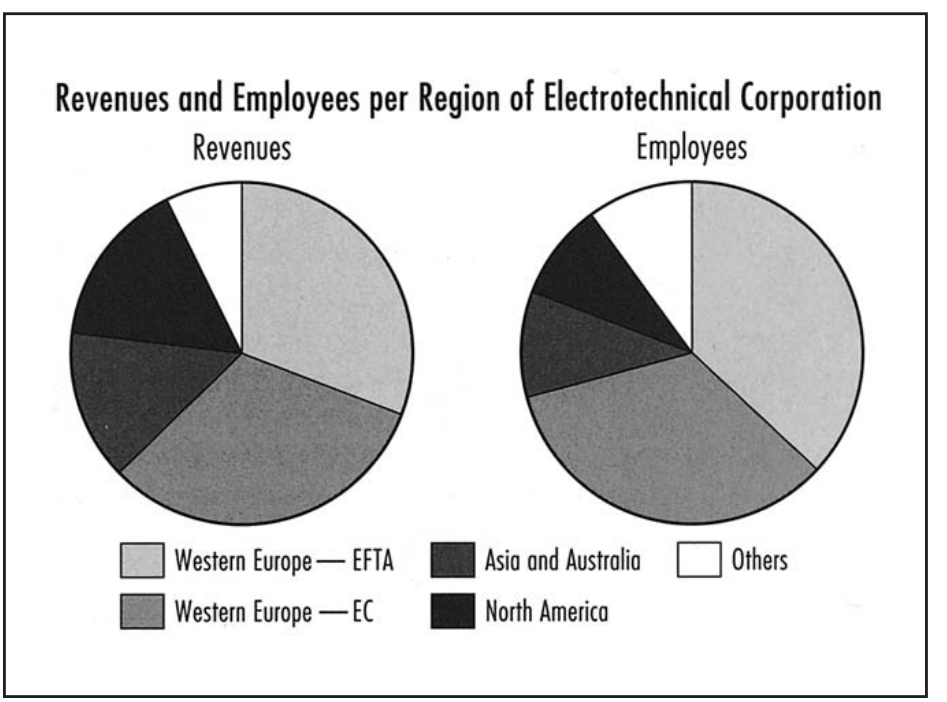


Figure 5.5. Corresponding wedges are hard to compare if, as in "Car Markets," they are not in corresponding positions.

Visual Tables: 8 More Legs to Stand On

We now turn to visual tables, which are pared-down graphs; objects are drawn so that their sizes or numbers vary in accordance with the amount being represented. Some visual tables use pictures of actual objects, and some use abstract content such as bars (without axes or scales).

1. Use a visual table to convey impressions of relative amounts.

If you want the audience to gain only a general sense of the ordering of measurements, a visual table is appropriate.

Example: Stacks of small water bottles, varying in number, may be enough to illustrate the increase in consumption over time.

2. Don't use the area of elements to convey precise quantities.

Because visual tables don't have scales, and because the visual system tends to distort our impression of the sizes of areas, don't use visual tables to depict precise amounts.

3. Provide labels to specify relevant precise amounts.

If precise values are important, label each content element with its amount (in the same way that you can label the sizes of individual pie wedges).

Example: If the numbers of balloons (presumably filled with hot air) indicate the number of speeches delivered by each of several politicians, you could draw a bunch of balloons for each pol, with a tag on the bottom that indicates the name of the politician and the actual number of speeches.

But be careful: This practice can easily result in a cluttered display, relegating the content elements to the role of mere decorations. If you want to convey precise numbers, consider simply using a tabular format instead.

4. Use sets of icons to provide labels or to compare relative amounts.

A special case of a visual table is the *isotype*, in which bars are created by repeating small pictures, each of which corresponds to a specific amount of the measured substance. These displays have two particular uses:

1. Pictures can provide a direct label of what's been measured, which reduces processing required to search for verbal labels.
2. Different pictures can indicate component parts of the measured entity.

Example: You could specify the number of male and female employees in the different divisions of a company (see Figure 5.6) by showing a column that contains copies of a male icon versus a column that contains copies of a female icon (perhaps of the sort that often grace the corresponding restroom doors) above a label for each division; each picture would stand for an increment of, say, five employees, with more pictures (forming a

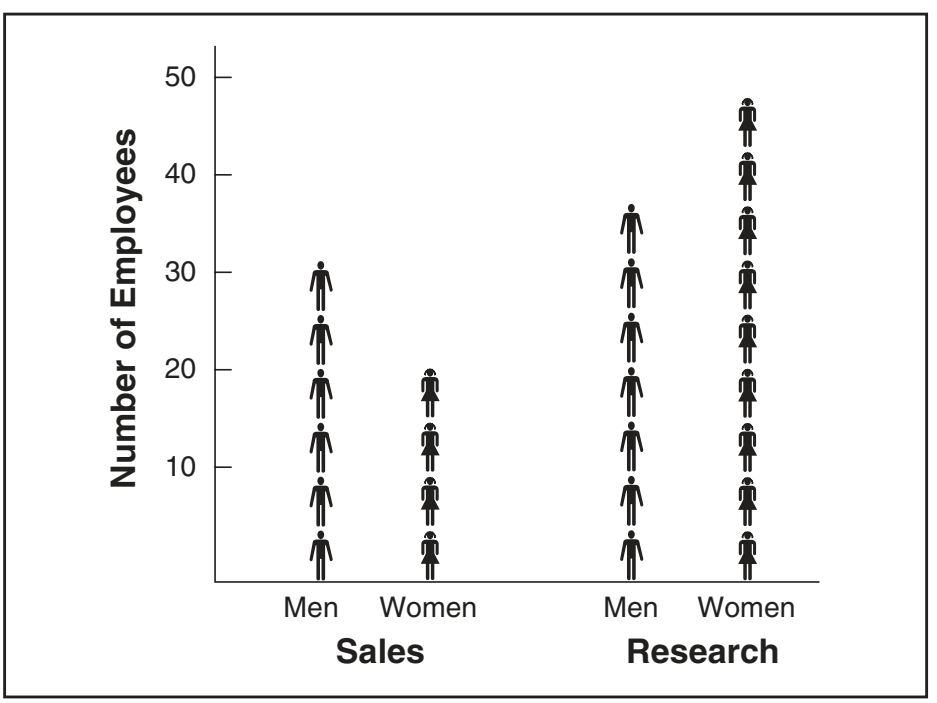


Figure 5.6. This bar graph could easily be turned into a visual table because the icons neatly label the bars, making the labels on the X-axis superfluous if the context in which the display is presented is clear enough. In this graph, each icon represents five employees, but the unit size is determined both by the ease of identifying the icons (if they are too small, they will not be discriminable) and the precision of the data.

higher stack) indicating more of that type of employee in that division. You can present precise amounts by showing a portion of an icon at the top of a stack, but be sensitive to how you truncate: A headless man at the top might lead audiences to draw the wrong conclusions about the quality of the staff! (In this case, make the top icon appropriately shorter by removing parts of the torso and legs—or just round off and only show intact icons, noting that you've done this in an explanatory note at the bottom or under the title.)

5. Make the appearance of the pattern compatible with what it symbolizes.

Don't fight human nature: Make the cover fit the book. If the properties of a pattern are not compatible with its meaning, the audience will have to work to understand the display. The visual impression of an amount or a difference should correspond to the actual amount or difference between the represented substances (Figure 5.7).

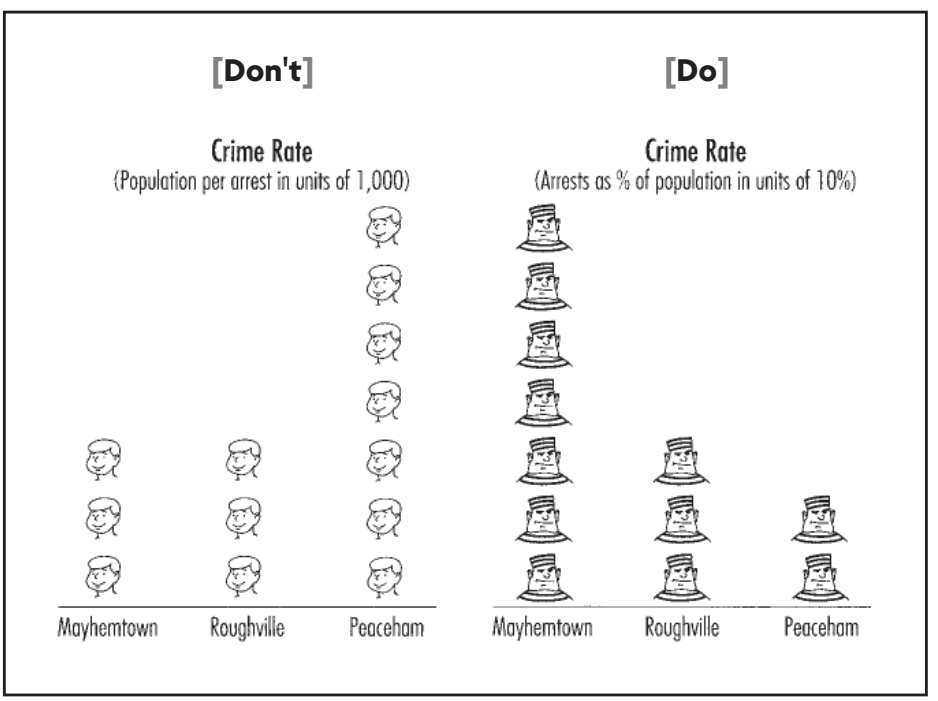


Figure 5.7. The icons should reflect the subject matter, which they do in both panels—but the appearance of the pattern is incompatible with the message in another way: "More" is seen as "more"—so longer bars or greater numbers of icons should specify larger amounts of the relevant substance or more frequent occurrences of the relevant event. In "Don't," greater numbers of icons actually indicate a lower crime rate! Do not convey "less" by "more."

6. Ensure that pictures used to illustrate measured entities depict those entities.

Pictures used to convey quantities either should depict the entities being represented, as in "Do" of Figure 5.8, or should have conventional symbolic interpretations that label those entities (such using an icon of an envelope for a post office or a donkey for the Democratic party).

7. Compare extents at the same orientation.

The "Don't" version of Figure 5.9 immediately demonstrates why this recommendation is a good idea: It is much harder to derive a relationship of quantities among the three elements when there is no common baseline and we must assess the absolute value of each in isolation, remember it, and compare it with the others. And, to make matters worse, our visual systems fall prey to an illusion that makes the same extent seem longer when it's vertical than when it's horizontal.

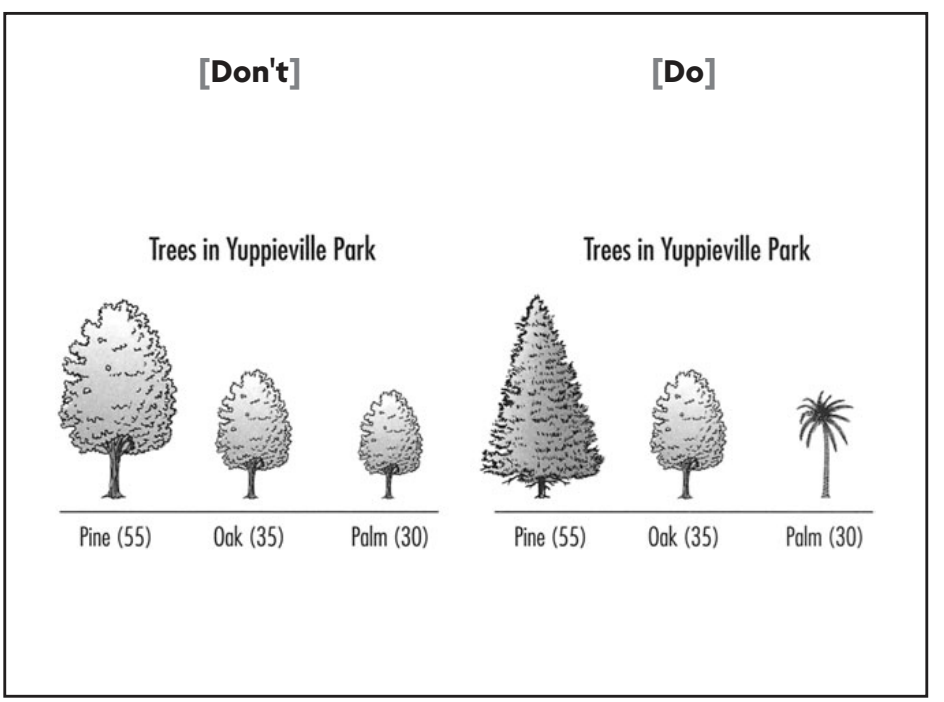


Figure 5.8. Pictures in a visual table should provide another kind of label and should not interfere with the interpretation of the display.

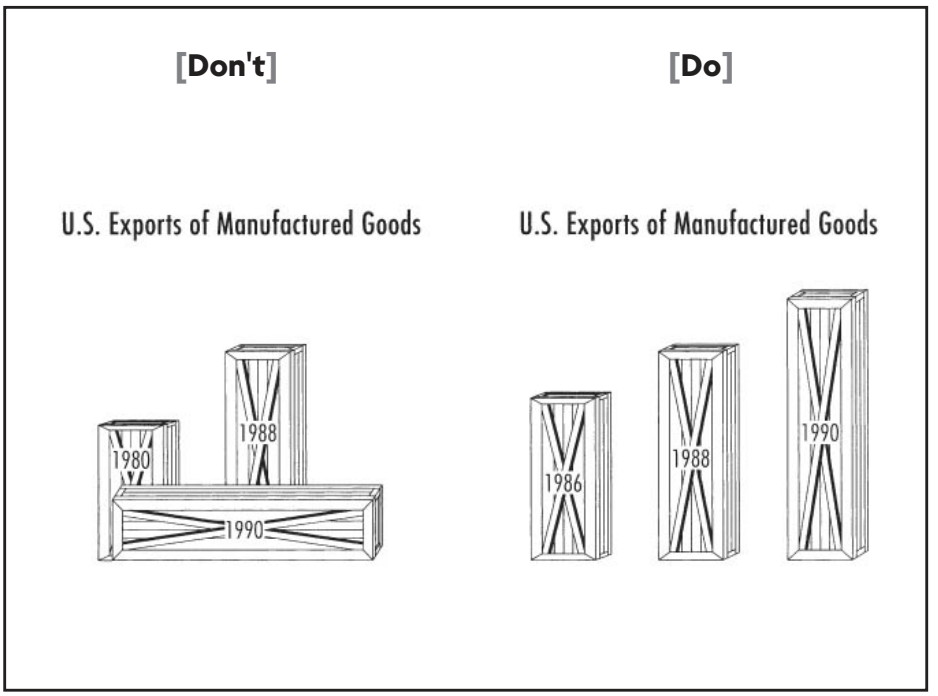


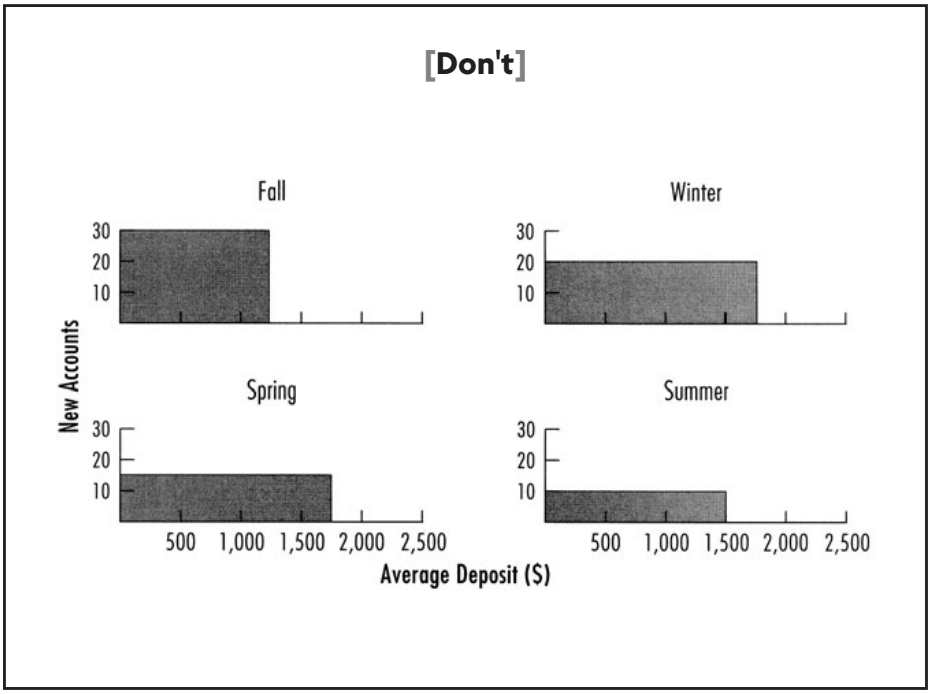
Figure 5.9. It is much easier to compare extents at the same orientation than at different orientations.

8. Don't vary height and width to specify separate variables.

A designer might use rectangular objects to present seasonal totals for new accounts opened at a bank. The height of each rectangle would indicate the number of accounts that were opened during that particular season; the width, the average amount deposited in such accounts; and the third variable, the area, would indicate the total revenue for each season. This scheme is shown in "Don't" of Figure 5.10. The problem: the audience members will see a rectangle—not height, width, and area as distinct elements.

Line Graphs: 11 Guidelines for Going in the Right Direction

Line graphs and bar graphs are the most common ways to illustrate quantitative information. This section describes when line graphs are more appropriate and summarizes recommendations for making effective line graphs.



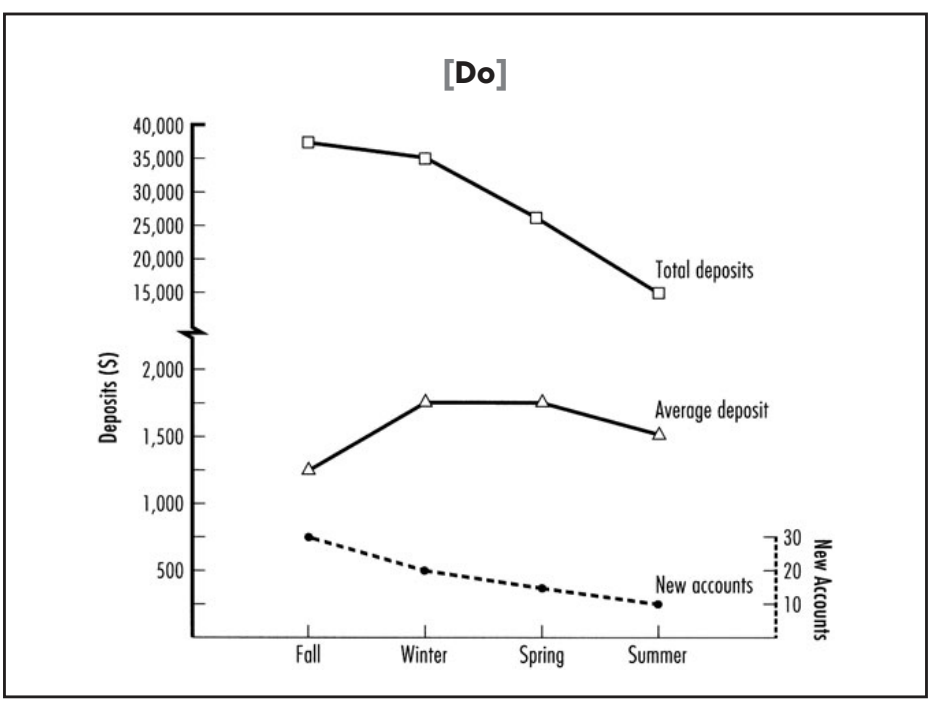


Figure 5.10. The line graph clearly indicates the number of new accounts, the average deposit, and the total deposits per season, as well as the relations among these measures. The same information is embedded in the four panels above, but considerable effort is required to decode it. We perceive height and width as specifying an area, not as separate variables.

1. Use a line graph if the X axis specifies a continuous scale.

The continuous rise and fall of a line is psychologically compatible with a scale that specifies the amounts along a continuous measurement scale. The yearly pattern of a young man's fancy is more clearly perceived from "Do" than from "Don't" in Figure 5.11.

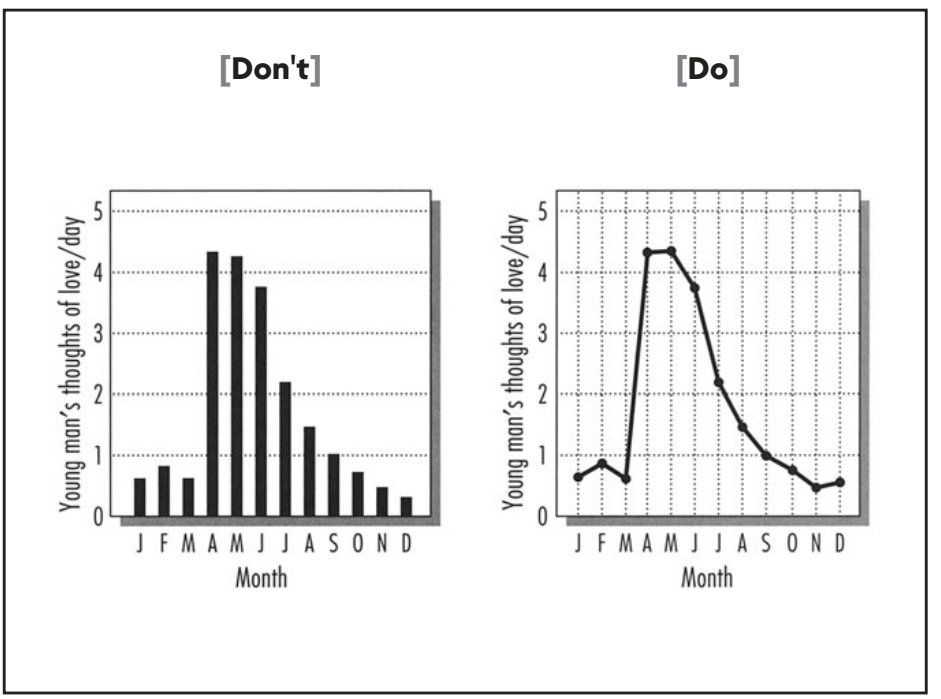


Figure 5.11. The continuous variation of a line is compatible with the continuous variation of time; if you want the viewer to note precise point values, put dots or symbols along the line.

2. Use a line graph to display trends.

The rise and fall of a line creates a shape, which we easily interpret as a trend—up, down, or a period of growth mixed with periods of contraction.

3. In general, use a line graph to display interactions.

One of the reasons it is so difficult to predict real-world events is that multiple variables interact, in a statistical sense: By "interact" I don't mean that two (or more) things have a personal relationship or that one necessarily changes another directly. Rather, I mean that

the effects of one variable can be understood only in the context of at least one other. When describing an interaction, you would say that the effects of X depend on the value of Y. For instance, the difference between the average temperature in June and January depends on whether you are in New York or Sydney. This interaction, between month and hemisphere, reflects the dependence of the value of one variable on the value of the other.

- If there is little chance that audience members will improperly infer a continuous scale along the X axis, then a line graph is a good way to convey interactions even if the X axis specifies rankings (not actual intervals) or even names, not amounts at all. But if a line graph would incorrectly imply a continuous scale on the X axis, use a bar graph.
- Except when a line graph would be misleading, avoid using a bar graph to illustrate interactions. In order to see bars as unified patterns that convey interactions, the viewer must mentally connect the ends of the bars, expending time and effort.

Example: Most experienced graph readers can immediately recognize that crossing lines indicate an interaction whereas parallel lines do not.

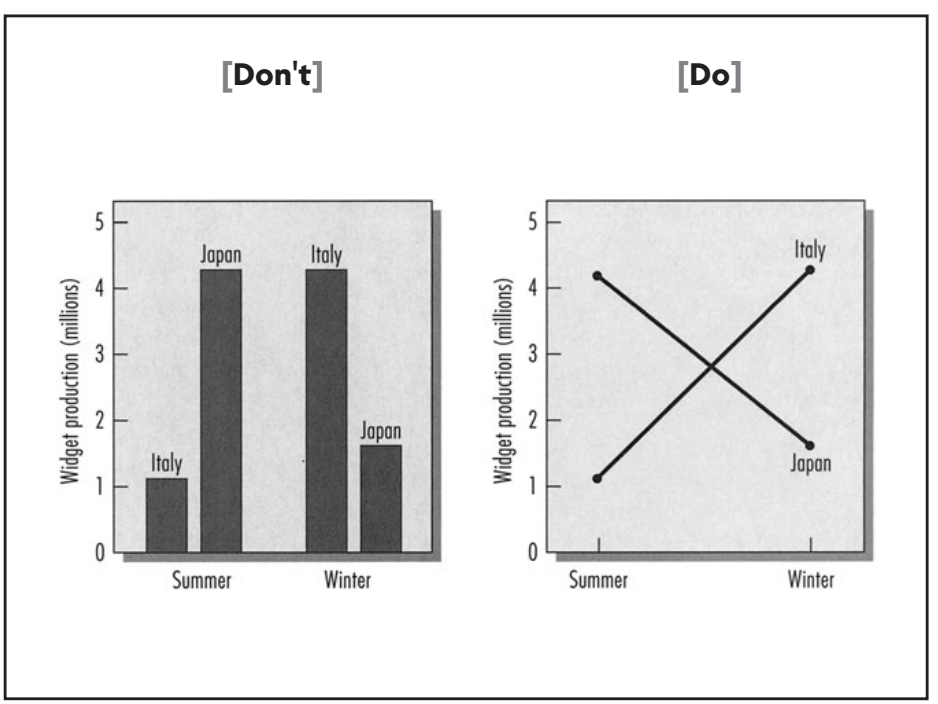


Figure 5.12. Experienced graph readers can interpret typical patterns of lines at a glance. Hence, it is good—especially in a presentation when viewers may not have much time to decode a display—to make use of familiar patterns to convey interactions.

4. Don't use a line graph to show point values.

If you use a line graph to show specific values, the audience members must locate and perceptually isolate a point along a line and note its height along the Y axis (Figure 5.13). This process violates the Principle of Limited Capacity; it requires the audience to work to understand you. According to one aphorism, "Work isn't a task, it's torture" (L. Bracken, http://deoxy.org/aaw.htm). Why make your audience work any harder than they have to?

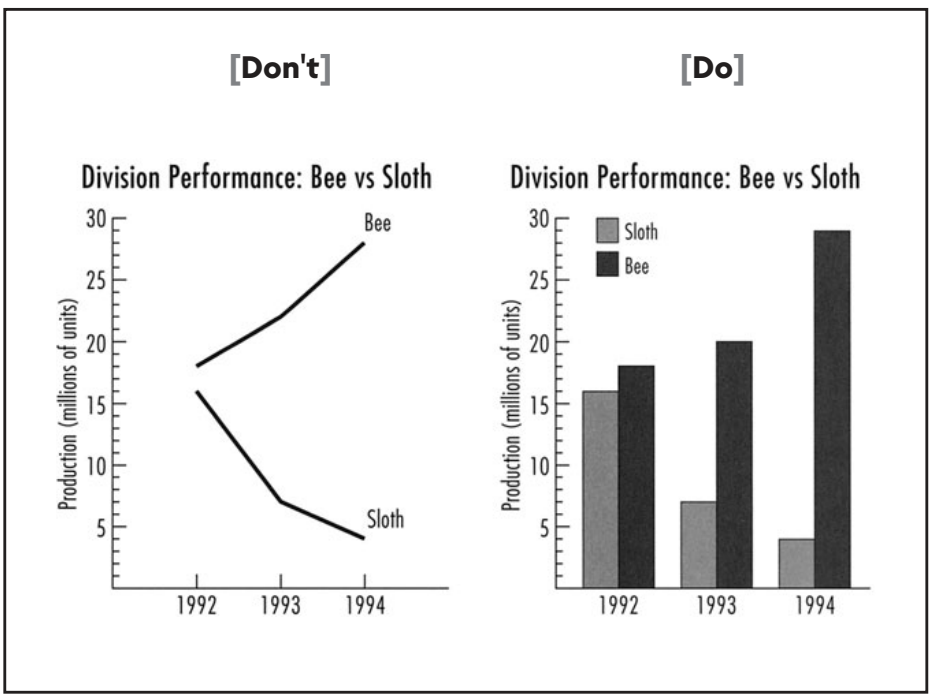


Figure 5.13. The heights of bars define specific points, whereas lines specify continuous variations. It is more difficult perceptually to break up a line into points than to detect the tops of bars.

5. Ensure that the X and Y axes are clearly identifiable and appropriately labeled.

The viewer should immediately be oriented to the X axis (which typically has values of what is varied or selected to be assessed) and the Y axis (which typically has values of what is measured). I was impressed by the following joke, not because it's funny—but because I've literally seen this happen! "The experimentalist comes running excitedly into the theorist's office, waving a graph taken off his latest experiment. `Hmmm,' says the theorist, `That's exactly where you'd expect to see that peak. Here's the reason (long logical explanation follows).' In the middle of it, the experimentalist says `Wait a

minute,' studies the chart for a second, and says, `Oops, this is upside down.' He fixes it. `Hmmm,' says the theorist, `you'd expect to see a dip in exactly that position. Here's the reason . . .' " (www.physlink.com/Fun/Jokes.cfm) Avoid such embarrassing moments by labeling the graph appropriately!

6. Vary the salience of lines to indicate relative importance.

Viewers will notice the most salient line first and interpret it as the most important. If you are looking at all three networks' morning shows and comparing their ratings, the graph on the left of Figure 5.14 is the style you want. But if your subject is NBC's "Today" and you want to emphasize that show's ratings against the other two shows in the field, the one on the right better illustrates your focus.

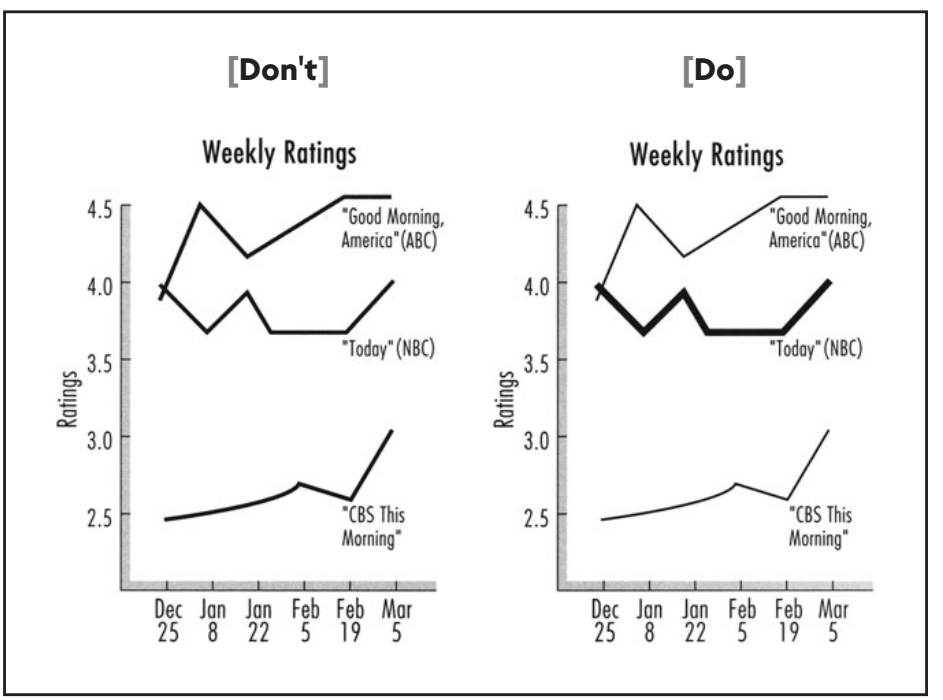


Figure 5.14. If the focus is on NBC, the graph on the right is preferable; the increased salience of the line for NBC immediately draws the viewer's attention.

7. Ensure that crossing or nearby lines are discriminable.

When lines are nearby or cross often, special care must be taken to ensure that they are discriminable. You can increase discriminability by

- using different colors;
- using dashed lines (and varying the dashes);
- in either case, making sure that the salience of the lines does not vary arbitrarily.

8. Ensure that dashes in lines differ by at least 2 to 1.

To be immediately discriminable from one another, dashed lines should differ in elements per inch in a ratio of at least 2 to 1 (this recommendation follows from the Principle of Perceptual Organization, specifically the aspect of Input Channels, explained in the Appendix). For example, if one line has 4 dashes to the inch, another line at least should have no more than 2 or 8 dashes to the inch.

9. If lines connect discrete points, make the points at least twice as thick as the line.

Some line graphs include dots or symbols that mark specific values on the X axis. In order to be easily discriminated from the line, the dots or symbols should be at least twice as thick as the line (Figure 5.15).

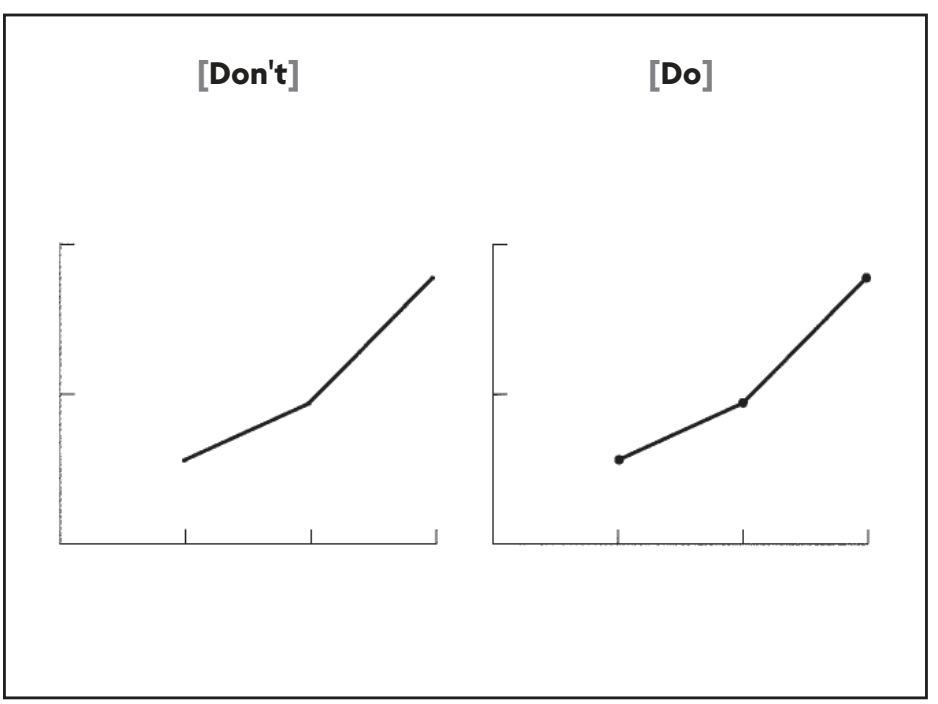


Figure 5.15. The points on the lines are especially important and should be emphasized by discriminable dots.

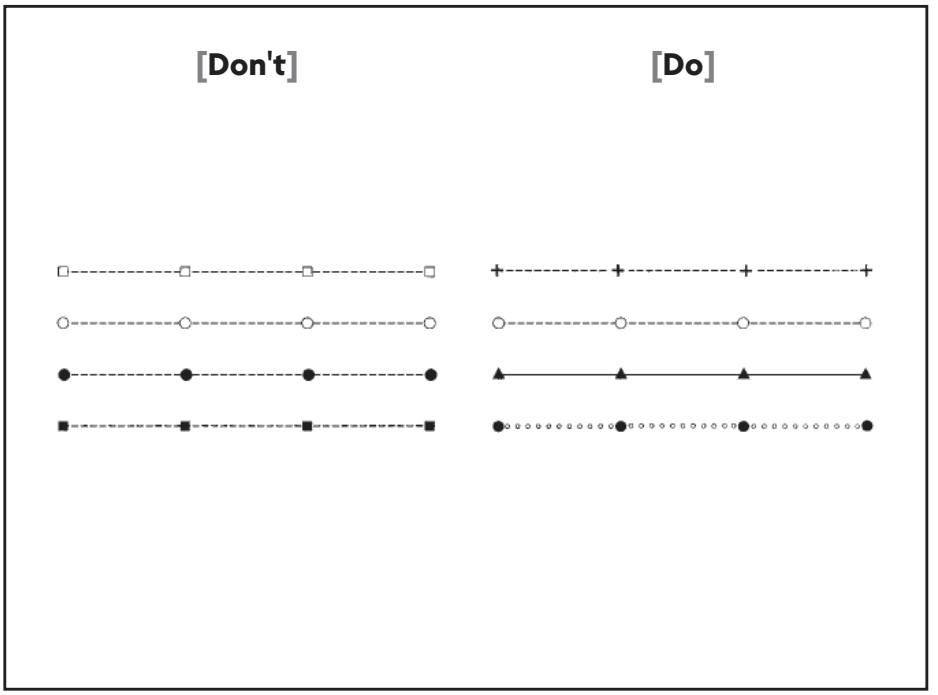


Figure 5.16. Studies have shown that the four lines and symbols on the right are highly discriminable.

10. Use discriminable symbols for points connected by different lines.

When lines are nearby or cross often, you can enhance discriminability by using visually distinct symbols for important points—either in conjunction with different dashed or colored lines or with solid lines.

Example: Plus signs, Xs, open circles, and filled triangles remain distinct even when reduced to relatively small sizes (Figure 5.16). However, ensure that the difference between filled and unfilled versions of the same element are easy to discern even when the graph is projected onto a screen.

11. Use an inner grid when precise values are important.

For all variants of line and bar graphs, include an inner grid if you want the audience to be able to extract one or more specific values, which can be read off the Y axis more easily if one can trace along grid lines. Especially with line graphs, insert heavier grid lines at regular intervals, which will help the viewer track from a point on the line to the axes (Figure 5.17).

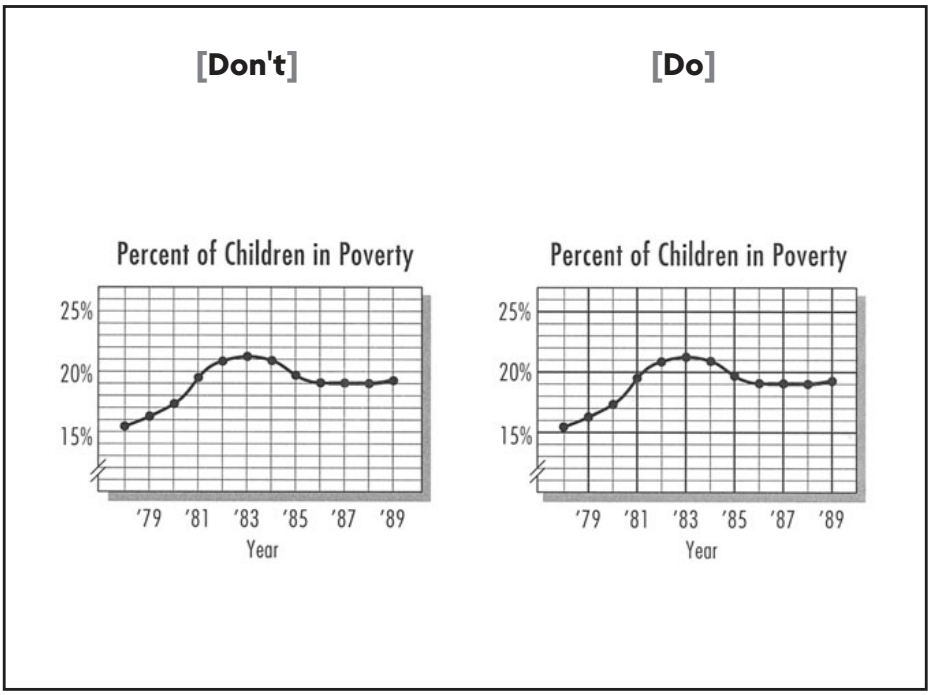


Figure 5.17. The staggered heavier grid lines help viewers to locate specific values on the axes for specific data points.

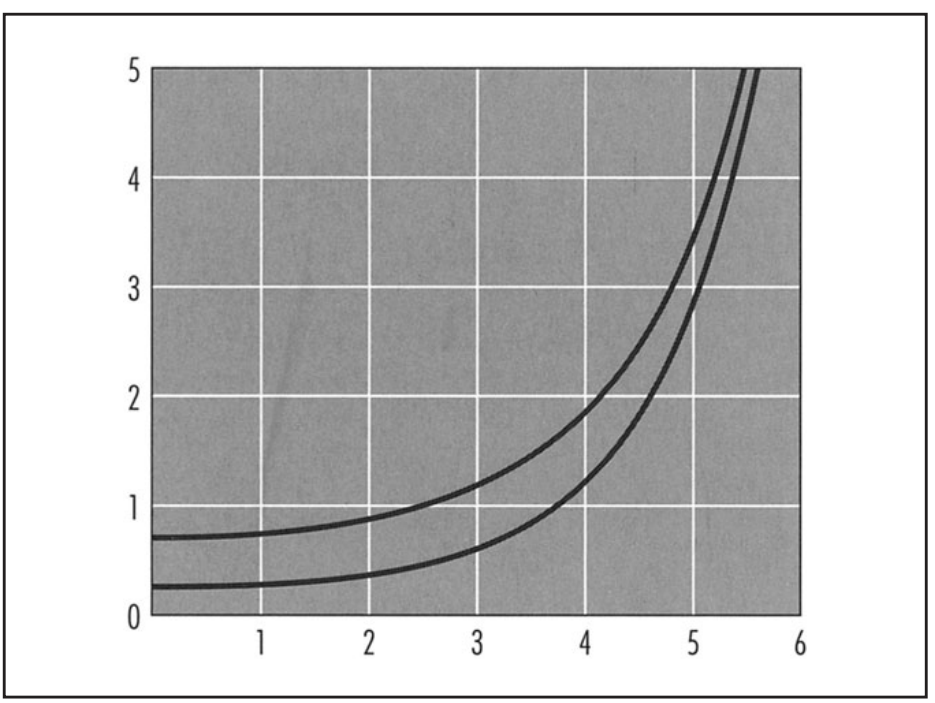


Figure 5.18. Without an inner grid, it is almost impossible to tell that the lines have the same difference in value on theY axis over the 2 and 5 values on the X axis.

Example: The line graph in Figure 5.18 illustrates a special case in which grid lines are particularly helpful. Look at the vertical difference between the lines over 5 and those over 2; which difference looks larger? In fact, they are the same distance apart, as you will see if you use the grid lines to compare the two differences. Our visual system tends to see the minimal distance between the lines, not their difference in height. Grid lines help the eye to focus on the vertical extent itself.

Bar Graphs: 10 Pillars to Build On

In this section we consider when to use bar graphs and crucial factors to keep in mind when presenting them.

1. Use a bar graph to illustrate differences between specific point values.

The ends of the bars in bar graphs specify particular values. If you want to contrast the measurements of a set of entities, use a bar graph.

2. Don't use a bar graph to illustrate trends.

To see a trend in a bar graph, viewers must mentally connect the tops of the bars. Why make them expend the effort? If showing trends is your goal, show them a line graph.

3. Use a bar graph if more than two values are on an X axis that does not show a continuous scale.

Look at the "Don't" version of Figure 5.19, in which the line makes it appear as if the income for consulting partners is accelerating rapidly. This visual impression is less striking in "Do". If the intervals of space along the X axis don't specify a continuously varying scale, a line will give a misleading impression of a trend. However, as noted earlier, if you want to show an interaction, consider using a line graph even in this situation. Your audience can quickly see that lines cross or diverge, but will need to work harder to see relations among sets of bars.

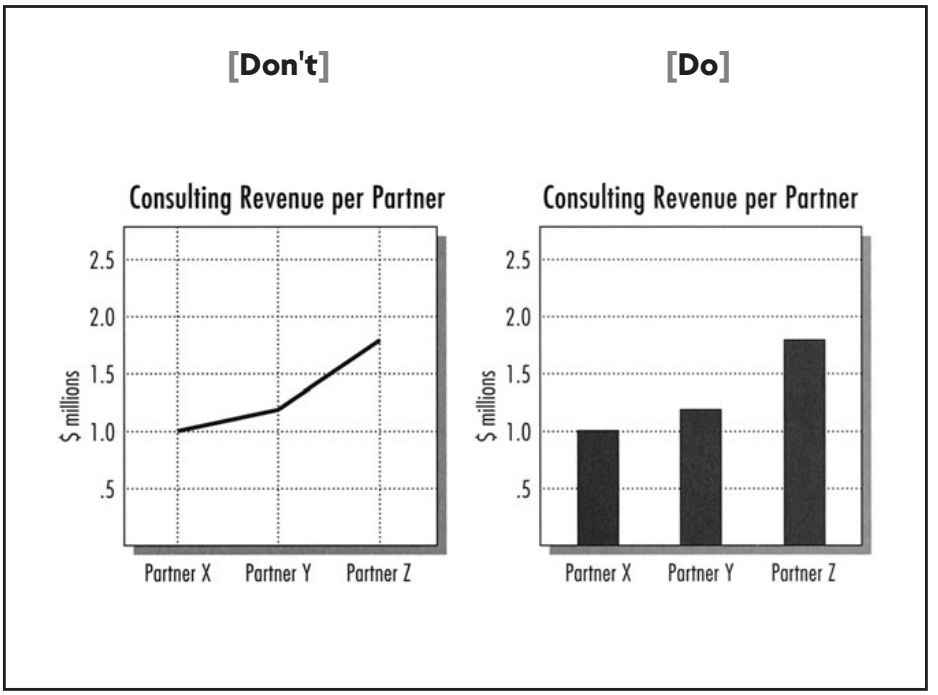


Figure 5.19. A line graph for these data inappropriately suggests a rapid rise along a continuous variation.

4. Use a horizontal bar graph if the labels are too long to fit under a vertical bar graph.

If a vertical bar graph requires novel abbreviations, audience members may receive less information than they need in order to decipher the material effortlessly.

As noted for line graphs, always be sure that the axes are clearly identifiable and clearly labeled. This is especially important for horizontal bar graphs, which are less common than vertical ones. The viewer should know immediately where to find the independent variable (things that are varied or selected to be assessed) and the dependent variable (the measurement values).

5. Don't use mixed bar/line displays to show interactions.

As is evident in "Don't" of Figure 5.20, it is more difficult to see interactions if a mixed display is used because bars and lines don't group to form simple patterns; to help the audience grasp interactions, use multiple lines.

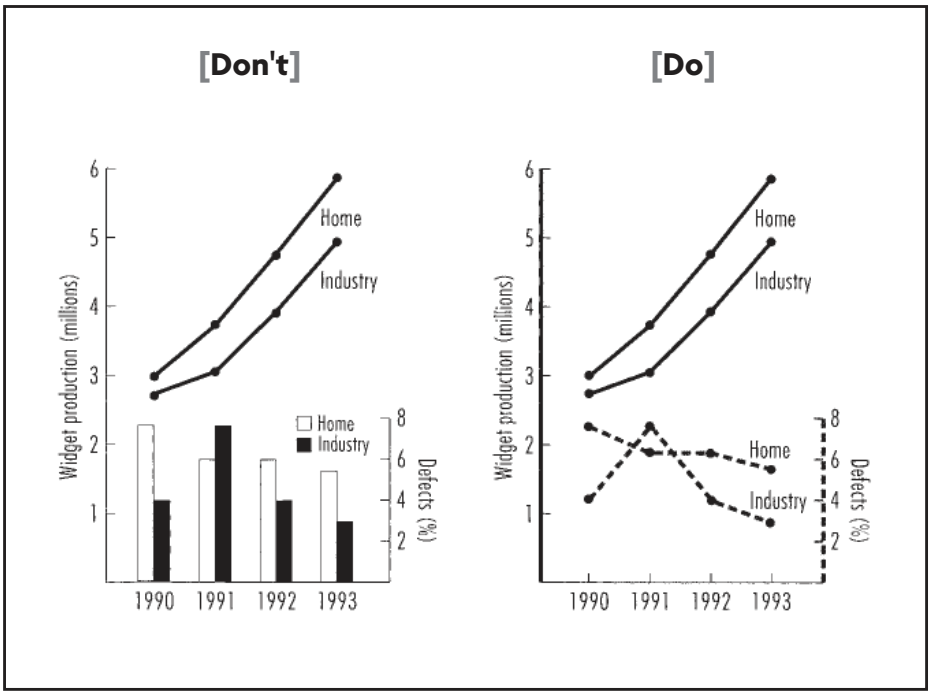


Figure 5.20. The trend toward decreasing numbers of defects with increasing production, especially for the industrial widget, is not immediately evident from the mixed line/bar display; the visual system more easily groups different lines into a pattern than it groups lines and bars.

6. Mark corresponding bars in the same way.

Mark the corresponding bars in each cluster the same way so that they are grouped appropriately (exploiting the grouping law of similarity), as in "Do". Try to compare the performance of the two brands of Coca-Cola in 1989 from the "Don't" graph in Figure 5.21. This is not as easy as it should be because, in a misplaced attempt at variety in the way the bars are shaded, the designer used different shadings for corresponding elements.

7. Arrange corresponding bars in the same way.

Say you want to graph numbers of young and old men and women voters for each of two counties. The demographic groups for each county would be on the X axis, and 4 bars, one for each of the groups (young men, old men, young women, old women) would be presented. The demographic groups could be indicated by different shading. The order of the bars for voter groups should be the same for each county, as in "Do" in Figure 5.22. If it isn't, the audience members will waste time trying to make comparisons and wondering why the order differs.

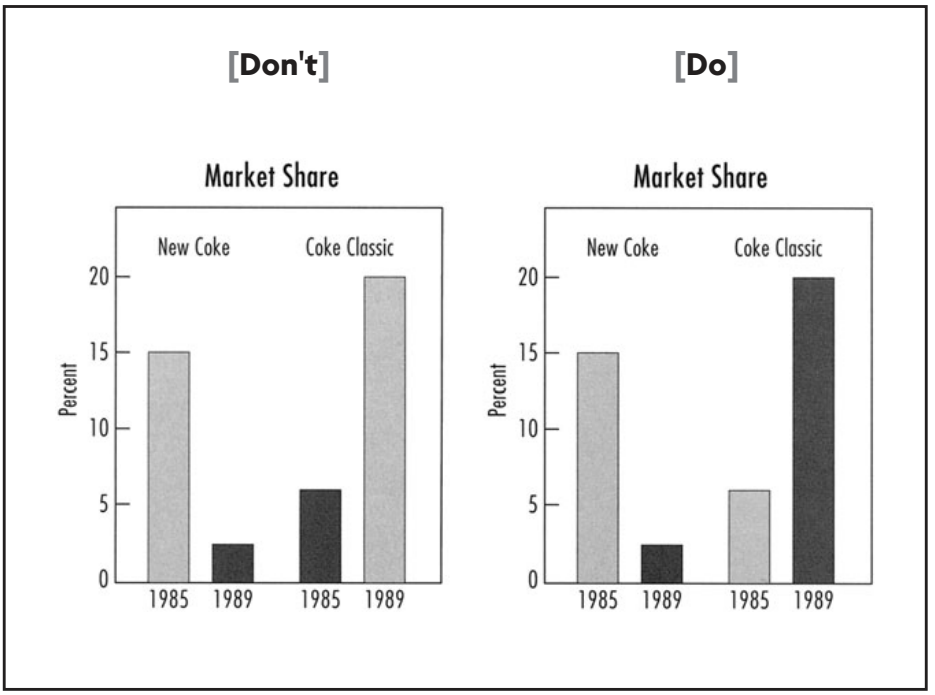


Figure 5.21. Inconsistent marking groups bars improperly in Don't, making it harder to compare corresponding elements.

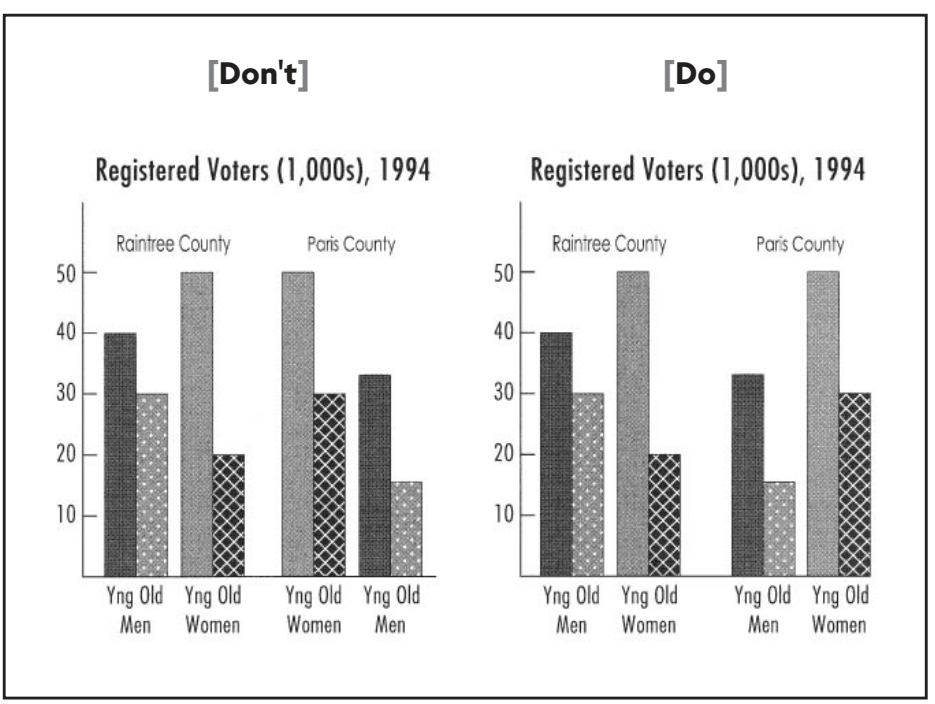


Figure 5.22. A gratuitous change of order makes it difficult to compare corresponding bars.

8. Don't vary the salience of individual bars arbitrarily.

Unless the emphasis is intentional, no bar should stand out from the others as happens in "Do" in Figure 5.23; if it does, the difference in salience will lead the audience to notice it first, and assume that it is more important than the other bars in the display.

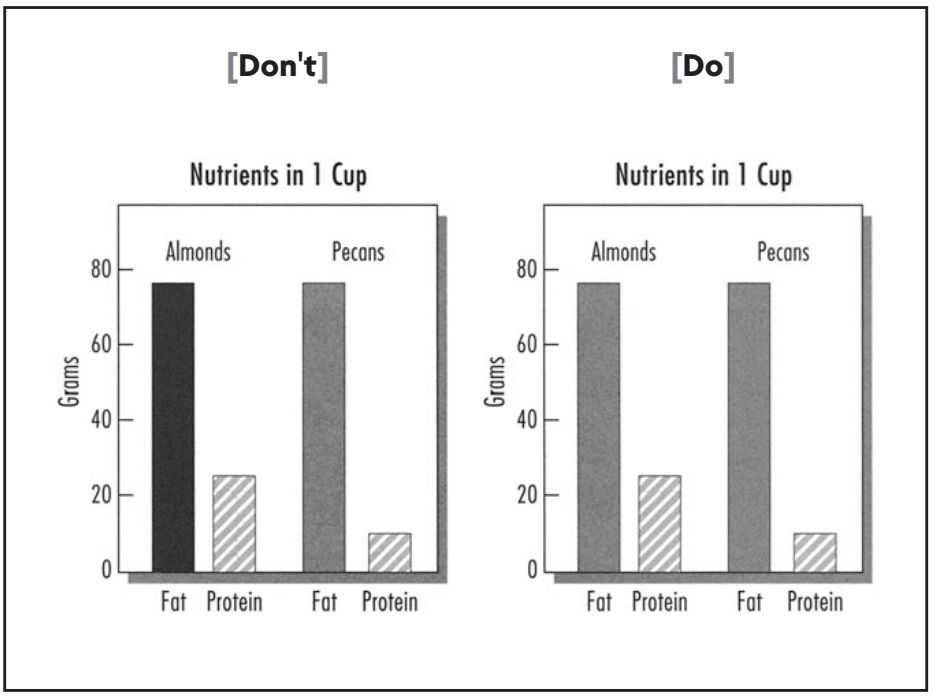


Figure 5.23. Making one element very salient will put it at the center of attention. If you want to emphasize that element, this is appropriate; but if you are simply trying to provide visual variety, as here, it is likely to confuse the audience.

9. Leave space between bar clusters.

Notice in Figure 5.24 how much easier it is to compare the market shares in 1985 for the three brands in "Don't" versus "Do". Group bar clusters over each appropriate location on the X axis and leave extra space between the clusters. As a rule of thumb, the space between clusters should be about as wide as two bars (this advice follows from the Principle of Perceptual Organization, specifically the aspect of Input Channels; see the Appendix).

10. Avoid extending bars beyond the end of the scale.

Don't make bars extend over the top of the Y axis, as in "Don't" in Figure 5.25 (or to the right of the X axis, in a horizontal bar graph) if

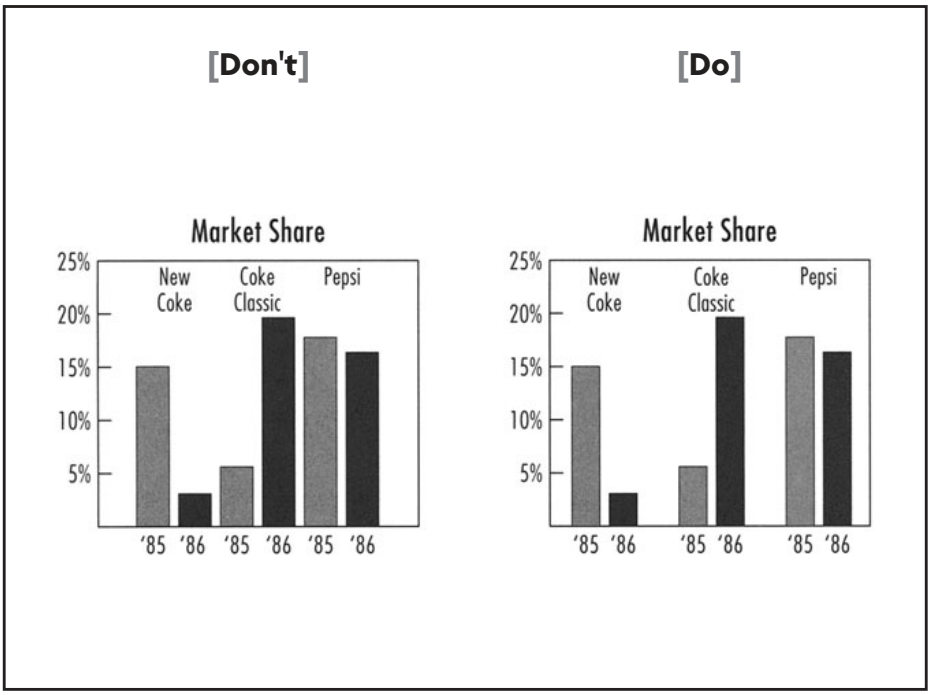


Figure 5.24. Six perceptual units are too many to apprehend immediately; proper grouping not only makes the display easier to take in but also helps to group the bars with their labels.

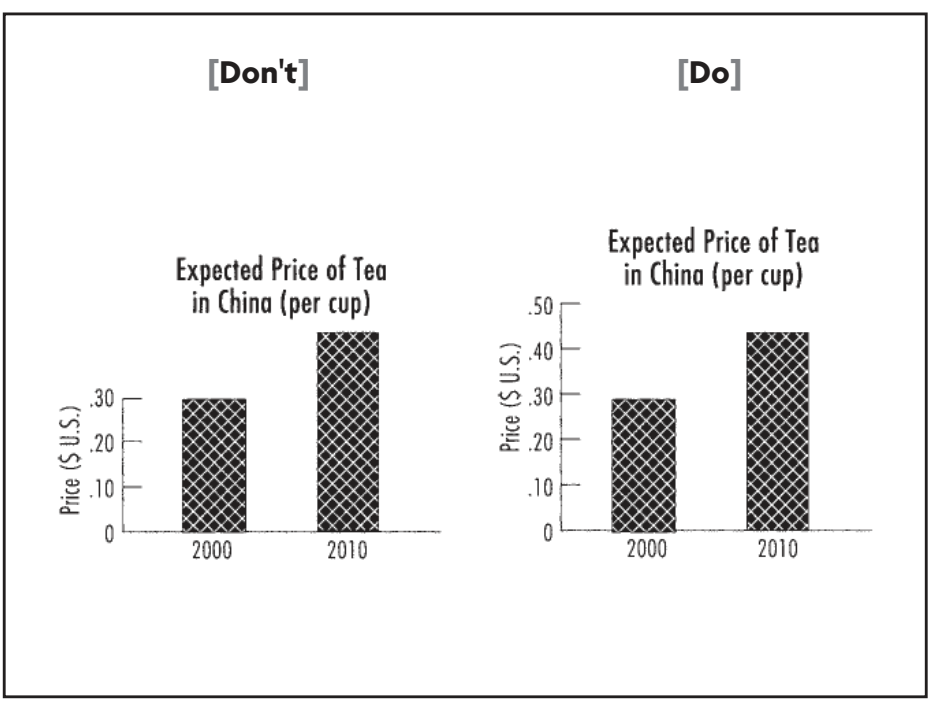


Figure 5.25. If the Y axis is too short ("Don't," at left), the viewer cannot immediately estimate the price of tea in 2010.

the viewers are supposed to be able to extract specific point values; mentally continuing the axis and its scale requires effort. However, if the point of the display is not to show specific values but only to indicate that, say, housing prices have shot through the roof, then you would just be adding extraneous details by including tick marks, labels on the Y axis, and so on. Some of the most effective graphs in *Time* magazine have tall bars that extend into the text, driving home the point that some trend has exploded beyond its usual boundaries.

Step Graphs: 4 Ways to Step Up

A step graph is a step-like line that can be thought of as formed by the tops of bars that have been pushed together.

1. Use a step graph to illustrate a trend among more than two entities that vary along a noncontinuous scale.

A step graph is useful if the audience members are supposed to notice relative changes over three or more values on the X axis but these values don't vary continuously and using lines would suggest inappropriate trends. Pushing the bars together creates a pattern that may allow the viewer to take in a trend at a glance.

- One drawback to these displays is that a bit more effort is required to isolate individual point values, and so a bar graph (where spaces between the bars isolate the individual bars, which convey point values) is more appropriate if this emphasis is your goal.
- Another drawback is that—depending on the data being plotted—patterns sometimes can be difficult to see. If this occurs, consider using a line graph (and in your presentation, explicitly point out the ways in which this graph may be misleading, but explain why it is good for showing the relevant interaction).

2. If the lines cross, don't include two or more lines in a step graph.

From the "Don't" graph in Figure 5.26 it is almost impossible to tell the percentage of automobile sales for the different countries. In these situations, present the data in separate panels, as shown in "Do". Such displays can illustrate different trends effectively, but do not illustrate interactions well.

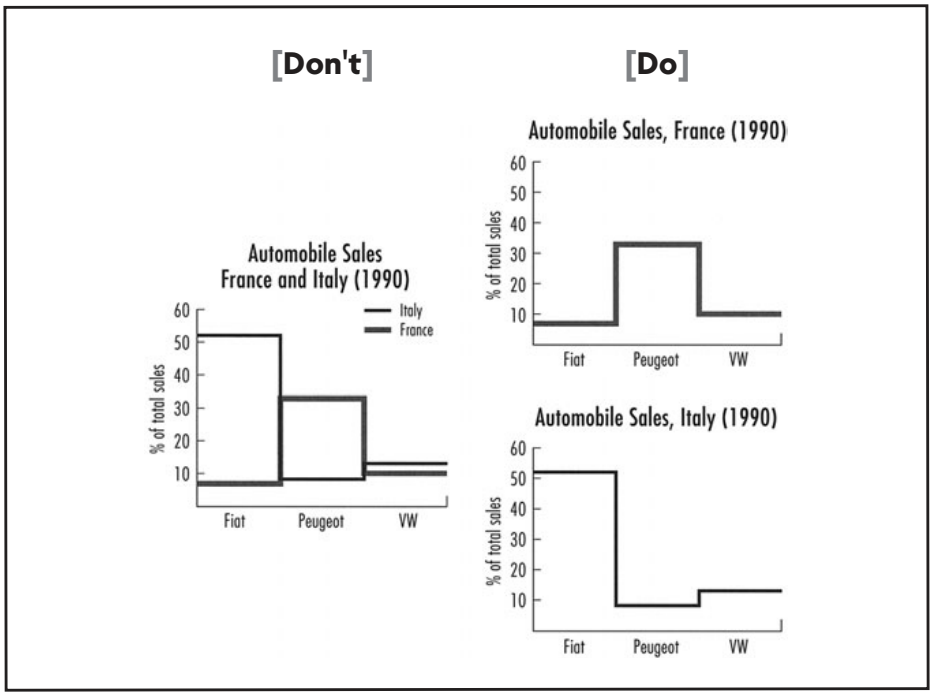


Figure 5.26. A combined step graph is difficult to read and easily mistaken as illustrating a cumulative total.

3. Make the steps of equal width.

In which version of the graph in Figure 5.27 does U.S. investment in Germany look more important? The reader will mistakenly perceive a wider step as representing a greater amount, even if that step is the same height as another, narrower step. Make sure the steps are of equal width, not drawn as in "Don't".

4. Fill the area under the line with a single pattern or color.

The primary virtue of step graphs is that they produce patterns that indicate specific trends. As is evident in "Do," if the area under the line is filled with a color or texture that is different from the background, the grouping law of similarity (as summarized in the Appendix) will lead the viewer to see that area as a single shape, facilitating recognition of a trend. But only a single color or texture should be used; otherwise, the impression of a single region will be disrupted.

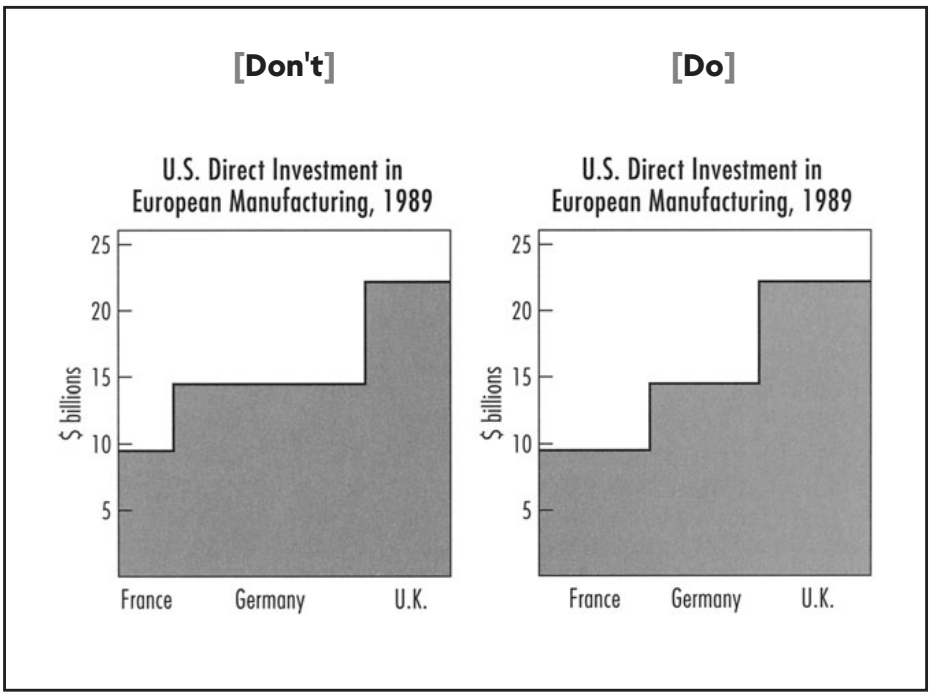


Figure 5.27. Making one step wider than the others gives it unequal importance.

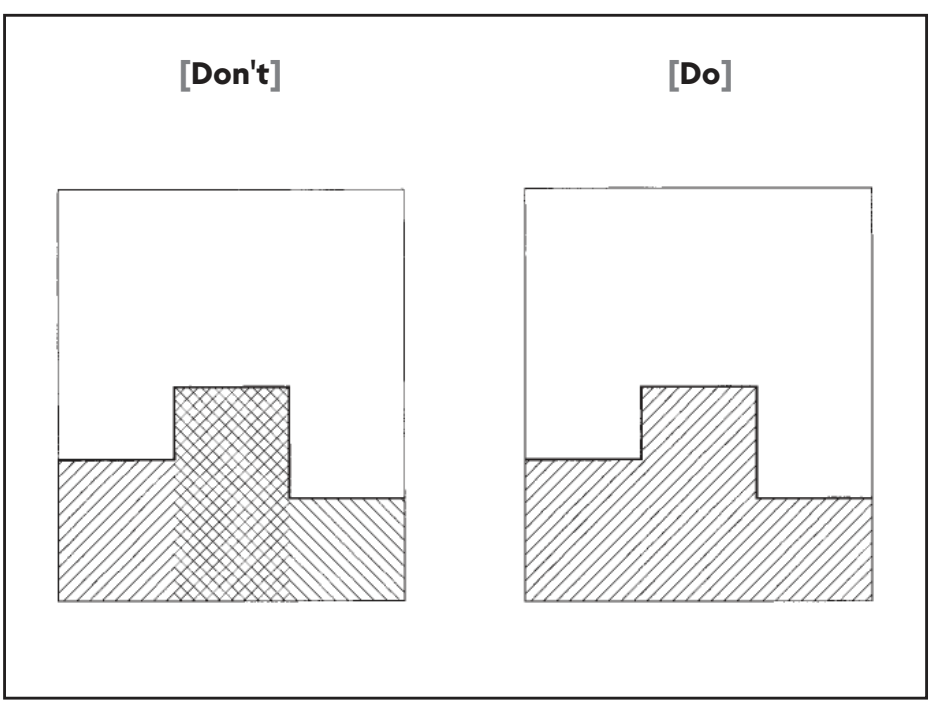


Figure 5.28. The shape of a series of rising and falling steps is meaningful and should not be disrupted.

Scatterplots: Ways to Avoid Scattershot

A scatterplot can look like random holes in a barn door or an army of ants marching in the same direction. This format uses the same sort of axes used in line and bar graphs, but now individual points are plotted rather than lines or bars.

1. Use a scatterplot to convey an overall impression of the relation between two variables.

Scatterplots are appropriate if you want audience members to obtain only an overall impression of the relation between two variables and you want the audience to have a sense of the variability of the relation between the variables.

Example: If the heights and weights of 100 people are plotted, viewers will see a cloud drifting up toward the right, indicating that the two variables are positively correlated. Moreover, viewers also can see that this correlation is not perfect; some very tall people are skinny, and some short ones rotund.

Don't include a grid, as in "Don't" in Figure 5.29; the point is to convey an overall trend, not individual values. Scatterplots often include far more points than can be grouped and seen as individuals; therefore, they are not a good idea if audience members are supposed to discern specific point values (use a table instead but present only the truly important values; otherwise you will overwhelm the viewers).

2. Avoid illustrating pairs of measurements for more than one category in a scatterplot.

Don't plot the pairs of measurements (such as the heights and weights) of different categories, such as Northerners and Southerners or Americans and Europeans, with different symbols for the points in the same scatterplot; the points will be intermixed and very difficult to discriminate. The only exceptions to this recommendation occur if you want to show that:

- data from two categories are intimately related (and so the fact that the points cannot be distinguished is itself compatible with the message being conveyed);
- data from two categories are clearly distinct (as evident when the clouds formed by different sets of points can be easily distinguished because they are in different parts of the display).

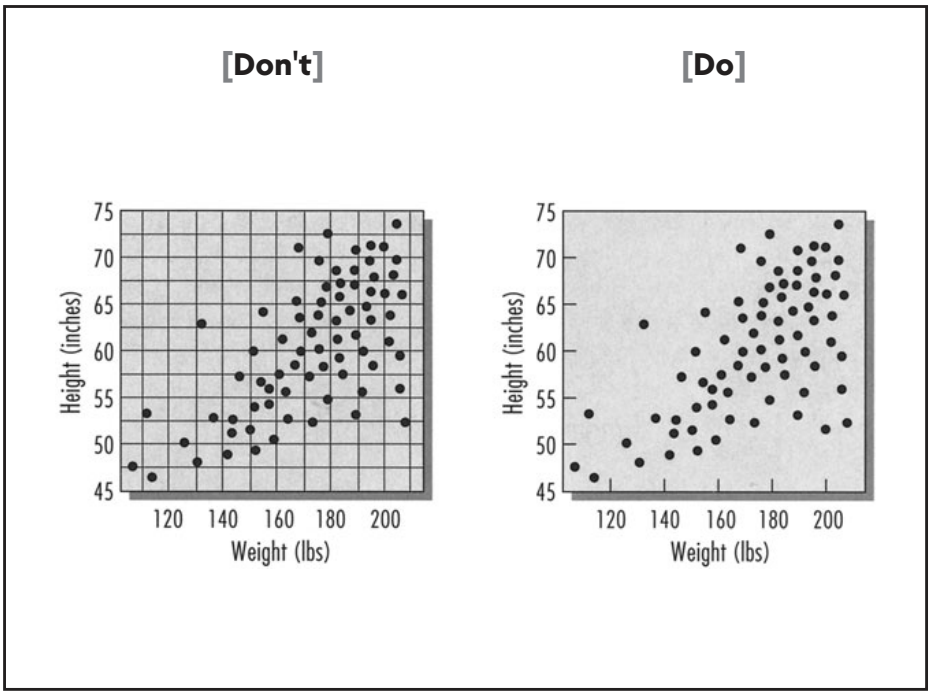


Figure 5.29. Scatterplots convey an impression of trends in the data; they generally are not well suited for conveying values of individual data points, and an inner grid is usually a distraction.

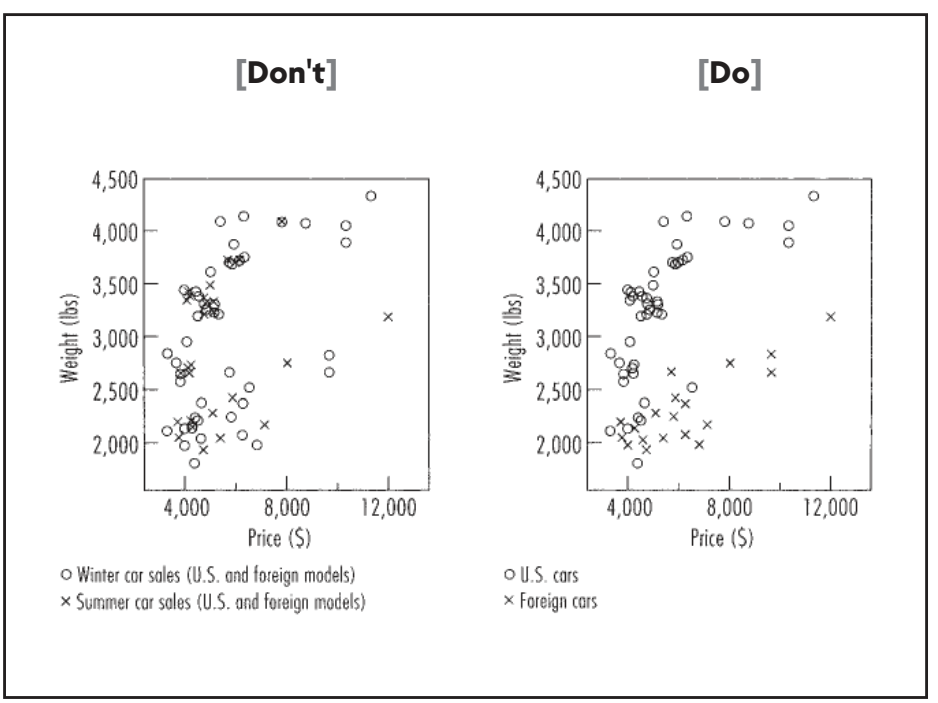


Figure 5.30. Measurements of different entities can be easily detected in a scatterplot only if their symbols occupy different parts of the display.

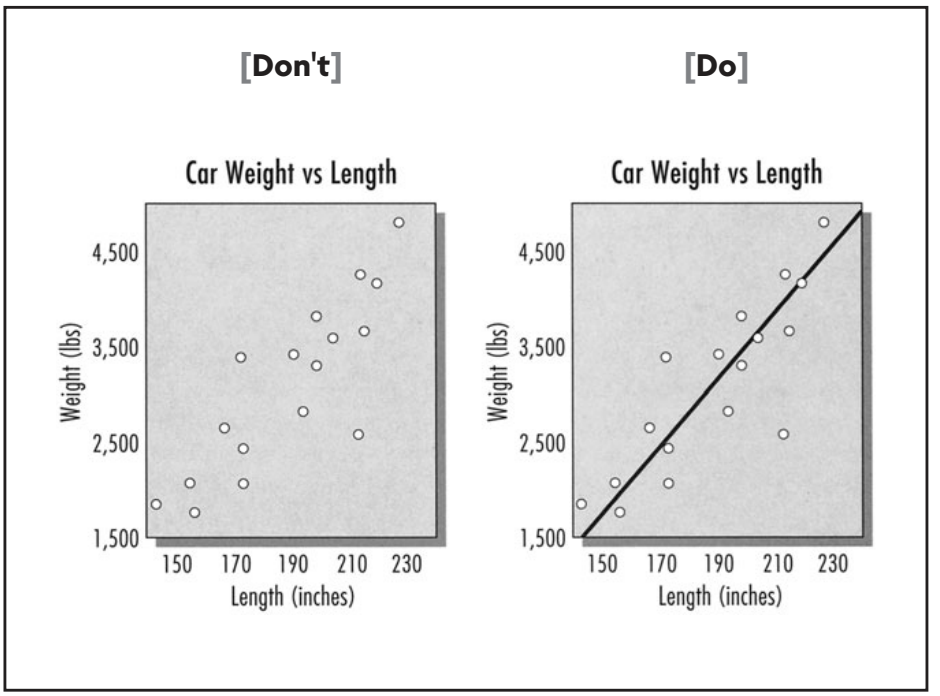


Figure 5.31. A best-fitting line serves as a summary of a trend and helps viewers to see the strength of the correlation between two variables.

3. Fit a line through a scatterplot to show how closely two variables are related.

In many cases, scatterplots are used to illustrate a trend, which can be summarized effectively by a line. Unlike the line in a line graph, this "fitted" line does not connect up sets of points. Rather, it falls through the "center"—the most densely populated area—of the cloud of points. The most popular way to fit such a line requires finding the location that minimizes the average distance of the points to the line (which is known as the "method of least squares").

Multiple Panels

When should you divide the data up and display it in separate panels?

1. Avoid presenting more than four perceptual units in one panel.

Our friend the *Rule of Four* applies here again: If at all possible, present no more than four bars or clusters of bars in a single panel. If there are more than four bars over each point on the X axis, or more than four groups of lines, find a way to divide the data into subsets and graph each subset in a separate panel (Figure 5.32).

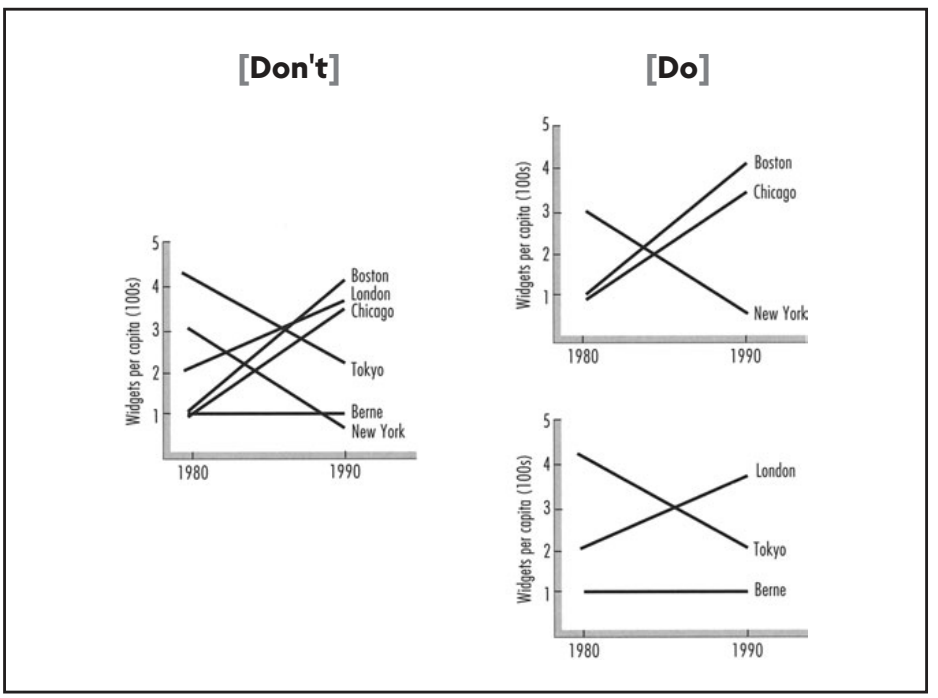


Figure 5.32. A display containing more than four perceptual units cannot be apprehended in a glance. Break such displays into separate panels.

2. Use multiple panels to highlight specific comparisons.

It is easier to compare material within a panel than across panels, so ensure that the data are organized to help the audience make the most important comparisons (Figure 5.33).

3. Graph different types of data in a single display only if they are strongly related and must be compared.

As a rule, don't graph different types of data in the same display; in most cases, graph the different data in separate panels of a multipanel

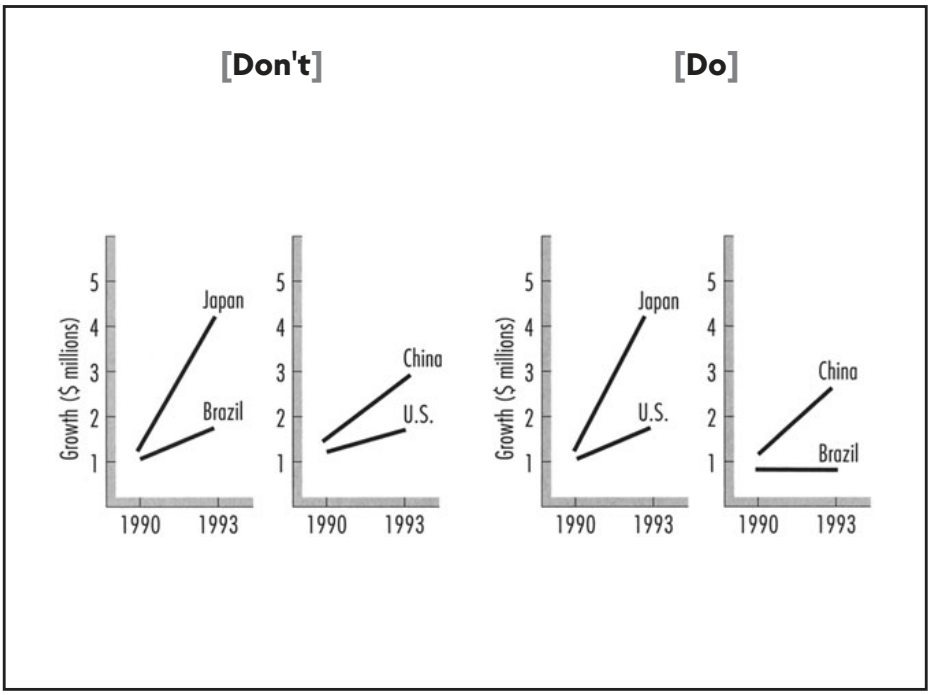


Figure 5.33. The intended comparisons—here, between two countries that have been industrialized for a long time and two emerging nations—are facilitated by appropriate grouping.

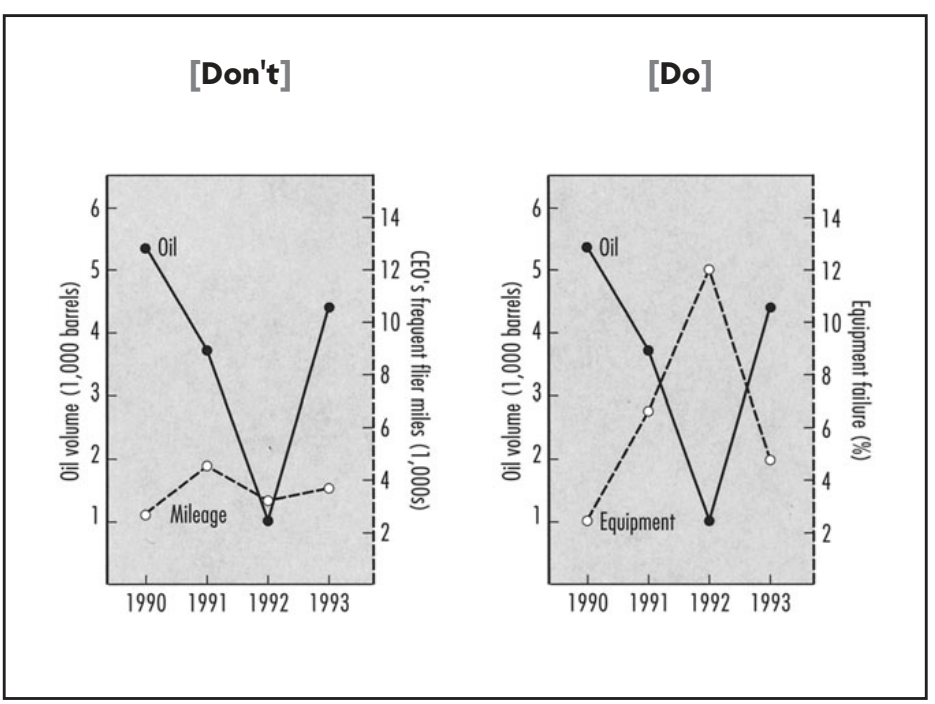


Figure 5.34. Plotting two related variables together can reveal useful information, such as the impact of equipment failure on the amount of oil that was pumped.

display. Graphing two types of measurements in the same display forces the audience to keep track of which bars or lines go with which scales, taxing their limited processing capacity.

The only exception to this recommendation occurs when the measured variables are intimately related and their interrelations are critical to the message being conveyed; in this situation, plotting the data in the same display allows the laws of perceptual grouping to organize them into a single pattern.

Example: The volume of oil pumped and the failure rate of a piece of pumping equipment are probably connected, and it makes sense to graph these two sets of measurements in one display, as in "Do" in Figure 5.34. On the other hand, oil volume and the CEO's frequent-flier mileage ("Don't") are not as immediately related, and the graph is not particularly helpful.

From the Principles to the Point

All of these recommendations stem from the eight psychological principles. If you understand the principles and how to apply them, graph design will become simple and natural. I again summarize each principle and note which ones contributed primarily to each of the major recommendations offered earlier in the chapter.

Principle of Relevance: Communication is most effective when neither too much nor too little information is presented. (We understand and remember a message more easily when the right amount of detail is used to make the point.)

- Present data needed for a specific purpose.
- Graph different types of data in a single display only if they are highly related and must be compared.

Principle of Appropriate Knowledge: Communication requires prior knowledge of pertinent concepts, jargon, and symbols. (We understand and remember a message more easily if it connects to what we already know.)

- Use concepts and display formats that are familiar to the audience.

Principle of Salience: Attention is drawn to large perceptible differences. (Big relative differences grab attention.)

- Use an exploded pie to emphasize a small proportion of parts.
- Explode a maximum of 25 percent of the wedges.
- Vary the salience of lines to indicate relative importance.
- Don't vary the salience of individual bars arbitrarily.

Principle of Discriminability: Two properties must differ by a large enough proportion or they will not be distinguished. (We need contrast to distinguish shapes, colors, or positions from each other and from the background.)

- Ensure that crossing or nearby lines are discriminable.
- Use discriminable symbols for points connected by different lines.
- Avoid illustrating pairs of measurements with different types of points for more than one category in a scatterplot.

Principle of Perceptual Organization: People automatically group elements into units, which they then attend to and remember. (These groups are easier to see and remember than the isolated components would be.)

- Compare extents at the same orientation.
- Don't vary height and width to specify separate variables.
- Ensure that dashes in lines differ by at least 2 to 1.
- If lines connect discrete points, make the points at least twice as thick as the line.
- Don't use mixed bar/line displays to show interactions.
- Leave space between bar clusters.
- Avoid extending bars beyond the end of the scale.
- Use an inner grid when precise values are important.
- Fill the area under the line in a step graph with a single pattern or color.

Principle of Compatibility: A message is easiest to understand if its form is compatible with its meaning. (For better or worse, the mind tends to judge a book by its cover.)

- Use a graph to illustrate relative amounts.
- Use a pie to convey approximate relative proportions.
- Use a visual table to convey impressions of relative amounts.
- Don't use the area of elements to convey precise quantities.
- Use sets of icons to provide labels or to compare relative amounts.
- Make the appearance of the pattern compatible with what it symbolizes.
- Ensure that pictures used to illustrate measured entities depict those entities.

- Use a line graph if the X axis specifies a continuous scale.
- Use a line graph to display trends.
- Use a line graph to display interactions over two entities on the X axis.
- Use a bar graph to illustrate differences between specific point values.
- Use a bar graph if more than two values are on an X axis that does not specify a continuous scale.
- Use a step graph to illustrate a trend among more than two entities that vary along a noncontinuous scale.
- Use a scatterplot to convey an overall impression of the relationship between two variables.

Principle of Informative Changes: People expect changes in properties to carry information. (And we expect every necessary piece of information to be indicated by a change in a perceptible property.)

- Label wedges of pie graphs if precise values are important.
- Ensure that the X and Y axes are clearly identifiable and appropriately labeled.
- Provide labels to specify relevant precise amounts.
- Use a horizontal bar graph if the labels are too long to fit under a vertical display.
- Mark corresponding bars in the same way.
- Arrange corresponding bars in the same way.
- Make steps in step graphs equal width.
- Fit a line through a scatterplot to show how closely two variables are related.

Principle of Capacity Limitations: People have a limited capacity to retain and to process information, and so will not understand a message if too much information must be retained or processed. (From a communicative point of view, less can be more!)

- Arrange wedges of pie graphs in a simple progression.
- Don't use a line graph to show point values.
- Don't use a bar graph to illustrate trends.
- If proportions vary greatly, don't use multiple pies to compare corresponding parts.
- Avoid presenting more than four perceptual units in one panel.
- Use multiple panels to highlight specific comparisons.